Experiment No.: 01

Title: DevOps Tools and Case Study

Batch: B–1 **Roll No.: 16010422234** **Experiment No.: 01**

Aim: To study DevOps tools and case study of a company.

Resources needed: Internet

Pre Requisite: Knowledge of DevOps concepts.

Theory:

What is DevOps?
DevOps is a culture which promotes collaboration between Development and Operations Team to deploy code to production faster in an automated & repeatable way. The word 'DevOps' is a combination of two words 'Development' and 'Operations'.

DevOps helps to increase an organization's speed to deliver applications and services. It allows organizations to serve their customers better and compete more strongly in the market.

In simple words, DevOps can be defined as an alignment of development and IT operations with better communication and collaboration.

Why is DevOps Needed?

- Before DevOps, the development and operation team worked in complete isolation.
- Testing and Deployment were isolated activities done after design-build. Hence they consumed more time than actual build cycles.
- Without using DevOps, team members are spending a large amount of their time in testing, deploying, and designing instead of building the project.
- Manual code deployment leads to human errors in production
- Coding & operation teams have their separate timelines and are not in sync causing further delays.

There is a demand to increase the rate of software delivery by business stakeholders.

CI/CD pipelines?
A **CI/CD pipeline** helps you automate steps in your software delivery process, such as initiating code builds, running automated tests, and deploying to a staging or production environment.

DevOps Automation Tools?
It is vital to automate all the software development processes and configure them to achieve speed and agility. This process is known as DevOps automation.

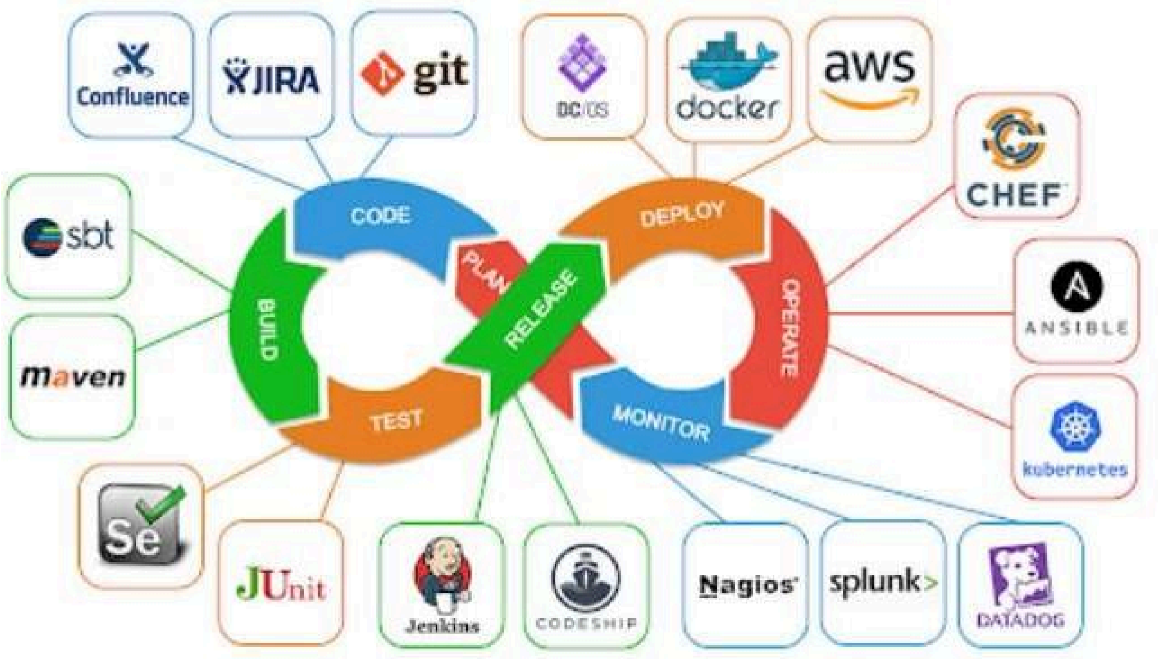

Figure 1: DevOps Processes and popular tools

Procedure: (group of 3-4 students, Zigsaw pedagogy)

1. Explore on the internet about DevOps and its tools.
2. Explore the internet and study any one case study/success story of DevOps.

Results: (Document based on concept understanding and presentation)

1. **Write in detail any two tools of each phase of the DevOps process.**

   I. Planning Phase Tools

   Tool 1: Jira
   - Purpose: Project management and issue tracking
   - Features: Sprint planning, backlog management, user story tracking, and workflow automation
   - Benefits: Helps teams plan sprints, track progress, and manage requirements efficiently
   - Integration: Seamlessly integrates with development tools and provides visibility into project status

   Tool 2: Confluence
   - Purpose: Documentation and knowledge management
   - Features: Collaborative documentation, requirement specifications, and team wikis
   - Benefits: Centralizes project documentation, facilitates knowledge sharing, and maintains project history
   - Integration: Works closely with Jira for comprehensive project management

   II. Development Phase Tools

   Tool 1: Git
   - Purpose: Version control and source code management
   - Features: Distributed version control, branching, merging, and collaboration

- Benefits: Enables parallel development, tracks code changes, and facilitates team collaboration
- Integration: Foundation for most CI/CD pipelines and development workflows

Tool 2: Visual Studio Code

- Purpose: Integrated Development Environment (IDE)
- Features: Code editing, debugging, extensions, and integrated terminal
- Benefits: Enhances developer productivity with intelligent code completion and debugging
- Integration: Supports multiple programming languages and integrates with Git and other tools

III. Build Phase Tools

Tool 1: Jenkins

- Purpose: Continuous Integration and build automation
- Features: Automated builds, pipeline orchestration, and plugin ecosystem
- Benefits: Automates build processes, reduces manual errors, and enables frequent integration
- Integration: Integrates with version control, testing tools, and deployment platforms

Tool 2: Maven

- Purpose: Build automation and dependency management
- Features: Project structure standardization, dependency resolution, and lifecycle management
- Benefits: Simplifies build processes, manages dependencies, and ensures consistent builds
- Integration: Works with IDEs, CI tools, and repository management systems

IV. Testing Phase Tools

Tool 1: Selenium

- Purpose: Automated web application testing
- Features: Cross-browser testing, test script automation, and web element interaction
- Benefits: Enables comprehensive UI testing, reduces testing time, and improves test coverage
- Integration: Integrates with CI/CD pipelines and testing frameworks

Tool 2: JUnit

- Purpose: Unit testing framework
- Features: Test case creation, assertions, and test execution
- Benefits: Facilitates test-driven development, ensures code quality, and enables automated testing
- Integration: Integrates with build tools and IDEs for seamless testing

V. Deployment Phase Tools

Tool 1: Docker

- Purpose: Containerization and application packaging
- Features: Container creation, image management, and environment consistency
- Benefits: Ensures consistent deployments, simplifies scaling, and improves resource utilization

- Integration: Works with orchestration tools and cloud platforms

Tool 2: Kubernetes

- Purpose: Container orchestration and management
- Features: Automatic scaling, service discovery, and load balancing
- Benefits: Manages containerized applications, ensures high availability, and automates deployment
- Integration: Integrates with cloud providers and monitoring tools

VI. Monitoring Phase Tools

Tool 1: Nagios

- Purpose: Infrastructure and application monitoring
- Features: System monitoring, alerting, and performance tracking
- Benefits: Provides real-time monitoring, prevents downtime, and ensures system health
- Integration: Integrates with various systems and provides comprehensive monitoring dashboard

Tool 2: Prometheus

- Purpose: Monitoring and alerting system
- Features: Time-series data collection, powerful query language, and alerting rules
- Benefits: Provides detailed metrics, enables proactive monitoring, and supports scalable architecture
- Integration: Integrates with Grafana for visualization and various exporters for data collection

2. **Write following questions based on your selected case study/success story of DevOps:**
    a. **Write the existing problem faced by the company.**

    Before adopting DevOps, Etsy faced several critical engineering challenges:

    - Slow and Risky Deployments: Code deployments were done manually once a week, often taking hours and frequently resulting in production outages.
    - Siloed Teams: Developers, operations, and QA worked in isolation, leading to poor collaboration and finger-pointing during incidents.
    - Low Deployment Confidence: There was no easy rollback process, and developers feared pushing new changes due to a lack of automated testing and monitoring.
    - Limited Innovation: The slow and error-prone release cycle hindered Etsy’s ability to deliver new features quickly to its users.

    b. **Which activities/problems are selected for DevOps solutions by the company?**

    Etsy focused on solving the following key activities/problems using DevOps:

    - Deployment Automation: They automated code deployment using internal tools like Deployinator to eliminate manual errors and reduce downtime.
    - Testing and Quality Assurance: Implemented automated unit, integration, regression, and smoke tests to ensure production-grade quality.

- Monitoring and Alerting: Built a strong observability framework using StatsD, Graphite, and Nagios to monitor real-time performance and trigger alerts.
- Infrastructure Management: Adopted Infrastructure as Code using Chef to manage environments consistently.
- Cultural Shift: Introduced blameless postmortems and gave developers end-to-end ownership of code from development to deployment and monitoring.

c. How has the company benefited?

Etsy saw several major benefits after implementing DevOps:

- Increased Deployment Frequency: From weekly to over 50 deployments per day.
- Faster Recovery: Significant reduction in mean time to detect and resolve issues.
- Improved System Reliability: Fewer incidents and quicker rollbacks ensured higher uptime.
- Greater Developer Productivity and Morale: Engineers felt empowered, trusted, and more confident in releasing code.
- Enhanced Customer Experience: Users received bug fixes and new features faster, improving overall platform satisfaction.

3. Share documents and present in the lab for all students.

CASE STUDY ON ETSY

Title: Etsy’s DevOps Transformation – From Fragile Deployments to Continuous Innovation

ABSTRACT

Etsy, a global online marketplace for handmade and vintage goods, encountered significant software engineering challenges as it scaled. The company struggled with slow, error-prone deployments, siloed teams, and frequent production outages that impacted user experience and developer productivity. To resolve these issues, Etsy adopted a DevOps approach starting in 2009. This case study delves into the strategies Etsy used—including cultural shifts, automation, continuous integration/deployment (CI/CD), and observability tools—to enable over 50 safe deployments a day. The transformation fostered collaboration, enhanced system reliability, and positioned Etsy as a DevOps pioneer in the tech industry.

INTRODUCTION

Etsy was launched in 2005 as an e-commerce platform designed to connect independent artisans with global buyers. By 2009, Etsy had grown rapidly but faced scalability and reliability issues due to outdated engineering practices. Software releases were infrequent, required considerable manual effort, and often led to service disruptions. Developers and operations teams worked in silos, causing friction and blame during system failures. Recognizing the need for agility, Etsy transitioned to a DevOps model that emphasized speed, collaboration, and resilience.

BACKGROUND AND SIGNIFICANCE

Etsy’s engineering challenges were not unique but represented a critical bottleneck in its ability to innovate. Deployments took hours, rollback processes were inefficient, and system outages were common. As Etsy grew, these issues became more pronounced, hindering its ability to deliver new features and maintain user trust. The company realized that solving these problems would require more than technical fixes—a holistic transformation in culture, practices, and tools was necessary. Etsy's journey became significant in the tech world as it demonstrated how a medium-sized company could adopt DevOps early and set a benchmark for others.

OBJECTIVES

- Streamline deployment processes to reduce time and failure rates.
- Foster collaboration between development, operations, and QA teams.
- Increase deployment frequency while maintaining system stability.
- Build a culture of shared responsibility and continuous improvement.

SCOPE

The DevOps transformation included cultural restructuring, CI/CD implementation, infrastructure automation, real-time monitoring, and developer enablement. The scope extended to all parts of the software delivery lifecycle, from code commit to production deployment and monitoring.

DEVOPS METHODOLOGY

Etsy’s DevOps methodology was grounded in the principles of agility, automation, and ownership. The company adopted continuous integration and deployment practices to ensure that code changes could be tested and pushed to production swiftly. Developers were given end-to-end ownership, including responsibility for production monitoring and incident response. Teams conducted blameless postmortems to analyze failures constructively. Frequent, small deployments minimized the risk associated with each change and facilitated rapid iteration.

IMPLEMENTATION METHODS

To support its DevOps transformation, Etsy built and customized several internal tools:

- Deployinator: A homegrown tool that allowed developers to deploy code with a single click, reducing complexity and dependency on operations staff.
- Try and Princess: Tools that automated pre-deployment testing and staging validation. These ensured that only production-ready code was pushed live, drastically reducing post-deployment issues.
- Feature Flags: Allowed for safer, incremental releases by controlling which user segments saw new features, minimizing potential disruptions.
- Infrastructure as Code (IaC): Chef scripts manage environment configurations, enabling consistency across dev, staging, and production environments.
- Monitoring Stack: StatsD and Graphite enabled custom metric collection and visualization, while Nagios handled alerting, creating a strong observability ecosystem.

The adoption of these methods was incremental and iterative, encouraging internal feedback and ongoing refinement.

ADVANTAGES

The DevOps initiative brought transformative advantages to Etsy:

- Deployment Velocity: Deployments increased from once per week to more than 50 per day, allowing faster delivery of features and bug fixes.
- Failure Recovery: Incident detection and resolution times significantly decreased, thanks to better monitoring and clear escalation processes.
- Developer Empowerment: Engineers gained control over deployment pipelines, fostering a sense of ownership and reducing friction with operations teams.
- System Stability: Smaller, isolated changes reduced the chances of system-wide failure and improved overall system uptime.
- Organizational Alignment: Cross-functional teams collaborated more effectively, leading to better alignment between business goals and engineering execution.

TYPES OF AUTOMATED TESTS

Etsy employed a layered testing strategy to ensure continuous code quality:

- Unit Tests: Automated checks on individual functions or components to ensure they work as expected in isolation.
- Integration Tests: Validated the communication between different services and system components to detect any interface issues.
- Smoke Tests: Basic tests executed after deployment to verify whether the critical functionalities of the application are working.
- Regression Tests: Ensured that new code changes didn't negatively affect existing features or logic.
- Monitoring Alerts as Tests: Real-time metrics and alert thresholds functioned as production-level tests, detecting anomalies post-deployment.

RESULTS

The results of Etsy's DevOps transformation were measurable and substantial:

- Deployment Time Reduced: What once took hours was reduced to minutes through automation and streamlined tooling.
- Deployment Frequency Increased: Moving from one deploy per week to 50+ daily enabled Etsy to stay ahead in feature delivery and bug fixing.
- Improved System Uptime: With better rollback mechanisms and continuous monitoring, outages became rare and less severe.
- Enhanced Developer Satisfaction: Developers experienced less friction, greater autonomy, and more confidence in releasing code.
- Customer Experience Boosted: Faster bug resolutions and new features enhanced platform reliability and user trust.

LEARNINGS

Etsy's journey offered several critical takeaways:

- Culture is Foundational: DevOps is most effective when supported by a culture of trust, openness, and shared responsibility.

- Incremental Change Works Best: Introducing DevOps practices gradually allowed teams to adapt and evolve effectively without resistance.
- Automation Pays Dividends: Investment in tooling and automated pipelines saved time, reduced human error, and improved reliability.
- Monitoring is Non-Negotiable: Real-time feedback through observability tools was essential to confidently deploy at high velocity.
- Failure is a Learning Opportunity: Etsy's adoption of blameless postmortems turned incidents into organizational learning experiences, rather than triggers for blame.

CHALLENGES

- Initial Resistance: Developers and ops had to adapt to new responsibilities
- Legacy Systems: Migration to automated and modular systems was gradual
- Tooling Overload: Building and maintaining internal tools took effort
- Balancing Speed with Stability: Rapid deploys needed constant monitoring
- Cultural Shift: Required executive support and mindset changes

SOLUTION

- Executive Buy-In: Leadership championed DevOps values and practices
- Incremental Implementation: Changes were introduced gradually with feedback loops
- Tooling Empowerment: Tools were designed to be lightweight and developer-friendly
- Cross-functional Teams: Reduced bottlenecks and improved collaboration
- Observability and Accountability: Empowered developers to own their changes end-to-end

CONCLUSION

Etsy's DevOps transformation is a prime example of how cultural realignment, supported by smart automation and continuous learning, can enable engineering excellence. By embracing blamelessness, developer ownership, and tooling tailored to internal needs, Etsy achieved high deployment velocity without compromising stability. This case study proves that DevOps isn't just a process—it's a philosophy that, when implemented effectively, drives innovation, scale, and resilience.

Questions:

1. Do DevOps have planning tools?
    a. No tools are available, it is done manually
    b. Many tools are available
2. Configuration management tool?
    a. Ansible
    b. Nagios
3. Continuous integration tool?
    a. Git
    b. Jenkins

Outcomes: CO1 – Understand the basic concept and stakes of DevOps.

Conclusion: (Conclusion to be based on the Results and outcomes achieved)
The experiment successfully demonstrated the role of DevOps tools in streamlining software delivery and improving collaboration across teams. The case study on Etsy highlighted how automation, CI/CD pipelines, and cultural changes drastically improved deployment speed and reliability. Overall, DevOps proved essential in achieving faster releases, reducing errors, and enhancing both developer productivity and customer satisfaction.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

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